# Speicherelemente und Flipflops

Übungen Digitales Design

**Lösung vs. Hinweise:**
Nicht alle hier gegebenen Antworten sind vollständige Lösungen. Einige dienen lediglich als Hinweise, um Ihnen bei der eigenständigen Lösungsfindung zu helfen. In anderen Fällen wird nur ein Teil der Lösung präsentiert.

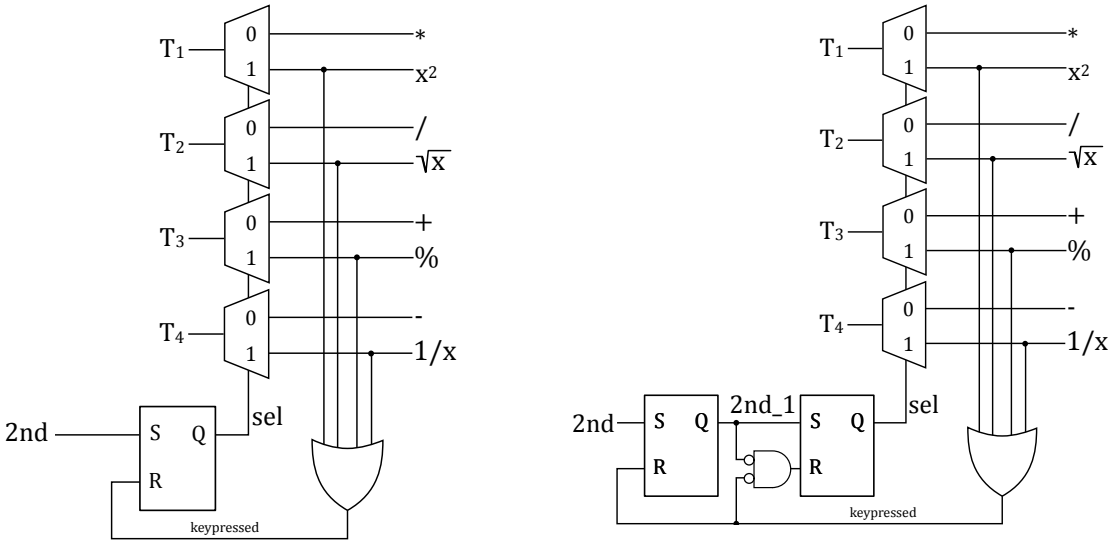
# 1 LAT - Speicherelemente

## 1.1 Anti-Prell-Schaltung

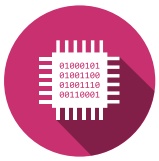
Switch on which one side is **`Reset`** and the other **`Set`**.

*lat/memory-01*

## 1.2 Tastenauswahl




*lat/memory-02*




## 1.3 Analyse eines Speicherelements

It is a SR latch with inverted inputs

*lat/memory-03*

## 1.4 Speicherelement

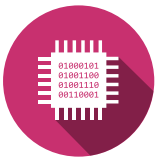
$$\begin{cases} g = 0 \Rightarrow s_n = 1 = r_n \Rightarrow \text{memorization} \\ g = 1 \Rightarrow \text{SR latch (set or reset)} \end{cases} \tag{1}$$

*lat/memory-04*

## 1.5 Synchronisation

Idea: Update the signal only at $\begin{cases} \text{clk=1} & \Rightarrow \text{D-Latch} \\ \text{rising_edge} & \Rightarrow \text{D-FF} \end{cases}$

*lat/memory-05*



# 2 LAT - Flipflops

## 2.1 Aufspürung von Übergängen

Using a D-FF as delay element and compare the 2 signals together (XOR-2).

*lat/flipflop-01*

## 2.2 Schieberegister

The output $Y$ is the same as the input $X$ with a delay of 4 clock period since there are 4 D-FF in the circuit.

*lat/flipflop-02*

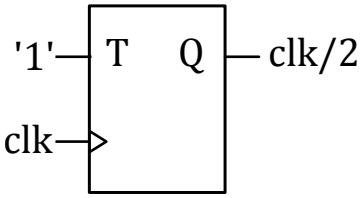
## 2.3 Flipflop, durch ihre charakteristische Gleichung bezeichnet

You need a D-FF and a multiplexer

*lat/flipflop-03*

## 2.4 Teiler durch 2

Only the solution for $\frac{\text{clk}}{2}$ is shown.




*lat/flipflop-04*

## 2.5 Ersatz eines Flipflop

### 2.5.0.1 Equation

$$\begin{aligned} E: Q^+ &= \overline{E}Q + ED \\ T: Q^+ &= T \oplus Q \Rightarrow \overline{T}Q + T\overline{Q} \Rightarrow T\overline{Q} + \overline{T}Q \end{aligned} \tag{2}$$

### 2.5.0.2 Table

| $T$ | $Q$ | $Q^+$ | $E$ | $D$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | - |
| | | | 1 | 0 |
| 0 | 1 | 1 | 0 | - |
| | | | 1 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | 0 |

Two variants are possible:

$$\begin{cases} E = 1,\ D = T \oplus Q \\ E = T,\ D = \overline{Q} \end{cases} \tag{3}$$

*lat/flipflop-05*

## 2.6 Schieberegister

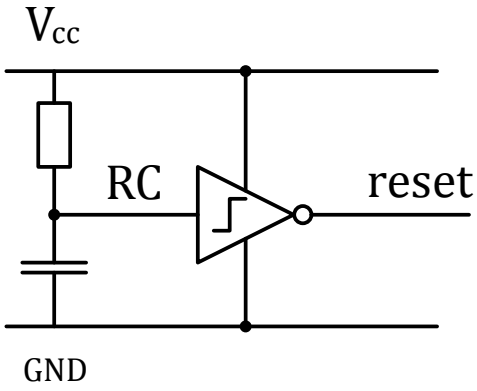
The real question is build a D-FF with a T-FF (see ex.Abschnitt 2.5 for the flipflop and ex.Abschnitt 2.2 for the shift register)

*lat/flipflop-06*

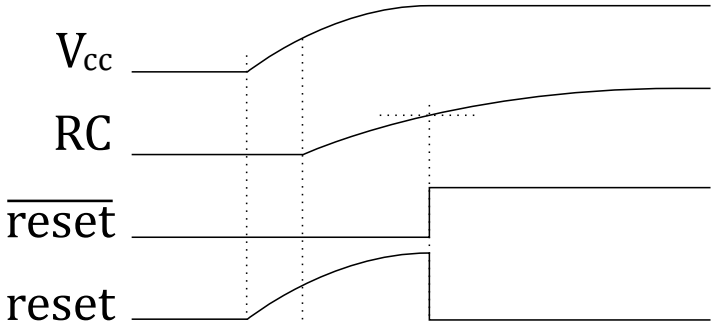
## 2.7 Asynchrone Nullsetzung

- RC-Circuit will be powere up with a delay
- Trigger converts the analog signal to a digital signal

Circuit

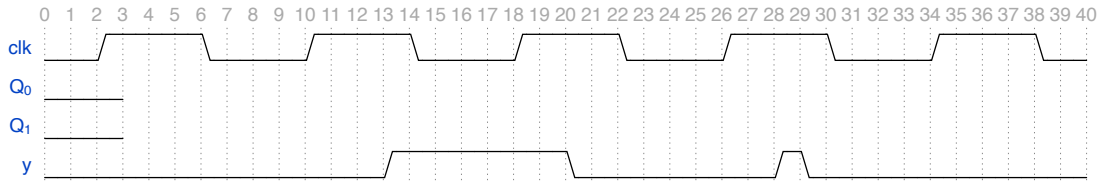



Schema

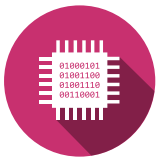



*lat/flipflop-07*

## 2.8 Asynchrone Schaltung




*lat/flipflop-08*


01000101
01001100
01001110
00110001